From Clock Synchronization to SOURCE Lensing and CMB

An Executable MCC/WRRA Research Record v1.0

Scope and current assessment

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This record consolidates the calculation sequence from event/time synchronization to finite SOURCE geometry, supplier energy and stress, actual CMB propagation, address-level calibration, finite mean preparation, lensing, and expanded observational evaluation. Original stage versions and failed controls are retained.

Validation inputs

Verified constants and observations, inherited finite address/slot/supplier ledgers, and declared frozen inputs. Calibration by known values is legitimate model construction. Calibration data and subsequent frozen-rule tests are distinguished.

WRRA transformation

State/event coding → frozen rules and resource/energy accounting → shared-carrier time/space readout → stress and physical evolution → observational evaluation. External physical modules are counted as integrated only where actually executed.

Outputs

Latest high-l quadratic form: 585.74750780. SOURCE deflection RMS: 0.00955659 arcsec. New numerical controls: 15/15; upstream reconstruction: 28/28. These are not a proof of the entire theory or a detection of SOURCE.

Refutation conditions

Revise a connection when held-out discrepancies, event/unit ambiguity, lost probability or ledger conservation, failed physical coupling, numerical nonconvergence, or observational incompatibility invalidate the relevant claim.

> Assessment: a conditional mean-stress model has an executed clock/carrier/supplier/CMB pathway. A unique universal shutter, complete underlying dynamics, and the individual-outcome/mean-gravity bridge remain open.

1. Time, phase, and decay-event cross-checks

Neutrinos: recalibration with one common mass scale

Retain integer modes (0,5,29), normal ordering and m1=0. Use Solar+KamLAND and Daya Bay uncertainty summaries to calibrate common q. Execute coherent vacuum relative-phase propagation.

Quantity	Output / meaning
q	1.735336790803 meV
(m1,m2,m3)	(0, 8.676683954, 50.324766933) meV
Mass sum	0.059001450887 eV
Central splitting residuals	Solar +0.37979%; atmospheric -0.35291%
First Pee minimum at 1 MeV	1.662008688 microseconds; calculated vacuum marker

Both splittings are compatible with the adopted marginal summary errors. A common q cannot match both centers exactly. PDG fits sharing experiments are sensitivity checks, not independent replications.

Hydrogen and the joint neutrino/hydrogen test

Execute the sequence $1/n^2 \rightarrow$ Coulomb/Dirac → residual S-state shift → fixed leading self-energy. The held-out 2S-8D5/2 residual falls from 7.880336 to 0.414264 MHz, about 19-fold. It remains outside experimental-summary precision. This truncation test is not a refutation of full QED.

Allowing q,A,L and common phase-rate G all to vary produces a rescaling degeneracy. With inherited q0 and an electron-based energy scale frozen, the conditional joint test gives G=0.999999994790629. Hydrogen dominates its precision; joining datasets does not uniquely determine a universal clock.

Muon-to-tau decay transfer

Calibrate a weak-rate scale with the muon mean lifetime and execute a positive finite event ledger. Adopted m^5, massive phase space and leading W corrections yield electron/muon partial-rate residuals -0.362612% / -0.418263%. Measured other-channel rates close the total tau lifetime.

> A phase cycle, a flavour-oscillation minimum, a mean lifetime, and an inverse partial rate are different observables. Shared units alone do not establish the identity of physical events.

2. Light/shutter and SOURCE geometry/supply

Conditional shutter with a 15-channel shared carrier

Inherited mode energy 22217.345682174 eV and phase resolution epsilon=0.05 set dt=epsilon*hbar/E. Known c sets dx=c*dt. Execute a shift on a finite 512-cell ring tensor the identity on 15 internal channels.

Quantity	Result
dt	1.481301966415863e-21 s
dx	0.444083157552045 pm
64-shift speed/c	Maximum relative discrepancy 6.23945e-14
Probability/channel/energy	Conserved to floating-point precision

c is an input. The ring is a computational boundary, not an inferred universe size. Reject direct-phase aliases at higher photon energies and test matched time/space refinement. Shutter conversion does not erase the earlier hydrogen residual.

First SOURCE metric witness

Read the actual 1,015,000 addresses and 500×29×70 finite index as a newly declared periodic graph. Execute the address bijection and dx edge spacing; the conditional 1024-point profile period is 0.454741153333 nm. Agreement with an independently derived SOURCE meter scale is not established.

Restoring residue ownership in supply bridge v0.2

A common total-slot kernel failed to reproduce local P/D/R allocation. Restoring the inherited residue-owned kernels with primes 2,3,5 reconstructs the actual address payload and local profiles. Preserve direct information-shape-to-density substitution as a negative control.

p_r(X)=0.5+1e-8[K_r(X/L)-1] → actual supply gap → energy/pressure modulation → seven harmonics. Include the supplier before exchange. Evolve all 28 real components in the GR subsystem.

> The executed 3D spectral carrier H=hbar*c*|k| is nonlocal in position. Strict microscopic causal support, Maxwell polarization/gauge dynamics and Lorentz completion are not claimed. The first macro wavenumber 0.05 Mpc^-1 is a calibrated input.

3. Actual SOURCE CMB propagation and diagnosis

Boltzmann execution

Propagate all seven SOURCE modes in patched CAMB 2.0.4 through photons, neutrinos and recombination. Adopt a regular radiation/CDM-isocurvature extension and constant delta_R. This is not a reverse continuation preserving the earlier late-epoch initial state.

In geodesic synchronous gauge: delta_R'=0; delta_D'=-k(z+v_D); v_D'=-H_conformal*v_D-k(rho_R/rho_D)delta_R. Include vacuum pressure perturbation and dark momentum in the metric. Disable the late-radiation approximation for SOURCE and retain late ISW.

First released-covariance evaluation

Frozen background branch	High-l quadratic form
Weighted	591.57034
Unweighted	1217.75123
Bins and calibration	TT/TE/EE: 613 bins; A_planck=1

Additional SOURCE TT peaks near 1.5e-5 microK². Agreement of the adopted baseline does not identify this small new contribution. Keep baseline calibration and SOURCE-mode propagation separate.

Units, modes and ledger-swap diagnosis

Control on the unweighted background	Quadratic form
Original	1217.75122
Replace dark ledger only	585.91392
Replace vacuum ledger only	1121.90312
Weighted original	591.57034

Check SI conversion, actual solver amplitude changes/superposition and momentum conservation. All 28 real trajectories remain finite. Logarithmic growth near density zero crossings is OPEN. Ledger swaps are counterfactual diagnostics, not newly generated upstream branches.

> Assessment: the dark background ledger is the stronger discrepancy driver in this configuration. The early-universe extension was already executed; generation/justification of its preparation is a separate question.

4. Implementing the correction at address level

Recompute from actual addresses

Within the inherited family e_s(n)w_n[1+lambda_s ln(n)/ln(N_U)], select lambda=(0,0.25,0) after the observational diagnosis. Recompute slots, residue, supplier and energy through CMB, without a density swap after readout.

Quantity	Output
Mean dark load	0.268 → 0.278925610976744
Additional dark energy	7.960139394787e-12 J/m³
Increment below address 100	Approximately 88.8% of total increment
H0	67.2568374662005 km/s/Mpc
ombh2 / omch2	0.022395806793 / 0.120378061774
Then-current CMB + SOURCE	585.9139188511556
Then-current numerical checks	41 passed

This is an executed upstream correction within the inherited family and a reality-consistency result of observationally informed model construction. Distinguish it from the original branch frozen before the comparison. No new SOURCE amplitude optimizer is introduced at each stage.

What the connection explains

The density correction can now be traced to particular addresses, residue and slots. Most of the increment is concentrated at low addresses, identifying a concrete modification site. Interpretation/applicability of the weight calibration remains a research question.

Refutation conditions

Reject or revise a connection if address/slot sums disagree, supplier energy or pressure fails to balance, the linear regime is exceeded, numerical refinement breaks the result, or observations disagree with the fixed candidate.

> 585.91 belongs to the new observationally informed address candidate. It does not mean the original unweighted branch achieved this agreement without further adjustment.

5. Finite mean preparation, lensing and likelihood

128-state finite phase preparation

Add a counted rho_phase=I/128 control register distinct from SOURCE mode |8>. Construct conditional supplier mean energy and mode covariance; compare with actual CAMB initial callbacks. Finite-phase/Fourier covariance relative discrepancy: 1.94218576587e-9.

Record initial control entropy ln(128). Check the regular retarded velocity in a near-radiation background series; the solver callback uses its regular leading term. No natural dynamical preparation or free repeated reset is proved.

Executed SOURCE covariance lensing

Compute T,E,phi at all l=2..6000 and the seven actual wavenumbers, and verify phi normalization directly. Do not fill SOURCE with a continuum Limber substitute. Include SOURCE phi in the adopted covariance remapping kernel. Positivity checks caught an int32 squared-normalization overflow; use float64 and recompute affected outputs.

Latest quantity	Result
SOURCE deflection RMS	0.009556588544 arcsec
Standard deflection RMS	146.804724052 arcsec
SOURCE/standard deflection variance	4.23765614476e-9
High-l quadratic form	585.747507795326
Low-T / low-E -2logL	23.425650359 / 396.117345700
CMB-marginalized lensing -2logL	9.066339240
SOURCE change in summed score	-8.07531250757e-7

Execute released Planck data with native Cobaya TT/EE, CMBlikes and Gaussian Plik-lite. Official clik and an unmarginalized full foreground fit are not executed. The offset-dependent sum near 1014.36 is not a universal chi-square with a simple degree count.

15/15 new controls and 28/28 upstream checks pass. The change 585.91 → 585.75 primarily reflects an extended baseline lensing calculation. SOURCE remains tiny; exact non-Gaussian remapping remains open.

6. Response scope and remaining physical links

Inputs → transformation → outputs → refutation

Calibrated finite supply/state → executed carrier/stress/CAMB/lensing → conditional CMB consistency → state, unit, conservation, convergence and observational tests. Numerical novelty alone is not the criterion for an explanatory achievement.

Response map for external cosmology feedback

Question	Current response
New mode in Boltzmann and Planck evaluation	Conditional native calculation executed
Dynamical stress/gravity/perturbation justification	Partial execution; foundations of adopted assumptions open
Initial spatial preparation and spectrum generation	Mean preparation constructed; dynamics open
Original branch without further adjustment	Discrepancy recorded; calibrated candidate separated
BAO likelihood / separate nucleosynthesis tests	Standard distances/PRIMAT outputs available; additional tests open

The earlier standard background includes BAO distances and PRIMAT interpolation: D/H=2.44177424096e-5 and helium mass fraction=0.24566720619. These are adopted-module outputs, not a new reaction-network run. Technical discussion with Professor Afshordi is context only; private email text and approval/collaboration claims are not included.

Next connection: individual outcomes and the mean

For one explicitly independent-address supplier-bit preparation, intrinsic relative RMS is D=0.34514, R=0.49415, versus mean phase modulation 1.21074e-8 / 9.48940e-9. Gravity currently reads ensemble-mean stress. Other correlations can change the variance; this is not an impossibility result for all preparations.

Construct counted supplier correlations or a finite averaging process accounting for independence, fresh resources, records and reset cost. The independent-frame diagnostic counts about 8.13e14 / 2.71e15 are not an implemented averaging mechanism.

Final assessment: an executed conditional clock-to-observables research record. Precision hydrogen holdout, unique clock/metric, preparation dynamics, individual-outcome gravity, and exact non-Gaussian SOURCE lensing are explicit next tasks.

Reproduction and source attribution

Preserve all original stage files. Store repeated large files once under their SHA256 and use prepare_workspace.py to restore exact original paths. No new observational fit or scientific recalculation is added by this consolidation task.

```
python prepare_workspace.py --verify-only
python prepare_workspace.py
pip install -r requirements.txt
python stages/neutrino_sync_v01/reproduce.py
python stages/cmb_preparation_lensing_v01/reproduce.py --verify-only
```

Each other stage has reproduce.py. Full CMB reproduction uses the bundled patched Linux runtime or gfortran/make rebuild, with heavy runs serial. Stage READMEs and frozen input JSON retain equations, assumptions and exact input provenance.

Primary physical and observational sources

Neutrino inputs: arXiv:2312.12907; arXiv:2211.14988; PDG 2026 neutrino-mixing review

Hydrogen inputs: arXiv:1107.3101; arXiv:1801.08816; arXiv:2111.08554; doi:10.1126/science.abc7776

Atomic/weak modules: NIST CODATA 2022; Rev. Mod. Phys. 80, 633; A. Pich arXiv:1310.7922; PDG 2026

SI and c: BIPM SI Brochure; c=299792458 m/s is a fixed input

CAMB: CAMB 2.0.4 source, included license and SOURCE patch; camb.readthedocs.io

Planck likelihoods: Planck 2018 V: arXiv:1907.12875; VIII: arXiv:1807.06210

Released native data: CobayaSampler/planck_native_data v1; planck_supp_data_and_covmats/lensing/2018

Native implementations: Cobaya 3.6.2; Plik-lite conventions by E. Calabrese and A. Lewis

Original finite-frame/supplier work: DOI 10.5281/zenodo.23329214. This v1.0 record consolidates subsequent calculations separately and does not overwrite that original record. Preserve third-party software/data credits and licenses.